

EP15 Measurement of the Acceleration Gravity, Using a Simple Pendulum

OBJECTIVE

To measure the acceleration gravity using a simple pendulum

THEORY

The periodic time T in seconds of a simple pendulum doing small vibrations can be shown to be given by

$$T = 2\pi \sqrt{\frac{L}{g}}$$

where l is the length in m, and g is the acceleration gravity in $m \cdot s^{-2}$

This equation transforms to

$$L = \left(\frac{g}{4\pi^2}\right)T^2$$

Thus a graph of L against T^2 will be a straight line and its gradient will be $K = \frac{g}{4\pi^2}$, then g can be obtained.

PROCEDURE

- (1) Rotate the screw at the top of the device to make the length of the string about 60cm.
- (2) Adjust the screws under the base of the device to make the frame vertical, and then the string is parallel to the frame.
- (3) Measure the diameter of the bob (denoted as d) five times along different direction using vernier caliper, and then calculate its average.
- (4) Measure the length of the string, denoted as l . So $L = l + \frac{d}{2}$.
- (5) Let the simple pendulum swing with a small angle. Record the time swinging 20 periods from the time tester. Repeat this 3 times.
- (6) Change the length of the string to make it respective about 70cm, 80cm, 90cm, 100cm, 110cm. Repeat procedure (5) for various lengths, obtaining six sets of data.

DATA RECORDING AND PROCESSING

Zero reading of the vernier caliper $d_0 =$

Table1 diameter of bob

n	1	2	3	4	5	average
d(m)						

Table2

n	Length in m			Time of 20 periods in s				Period T in s	T ² in s ²
	String <i>l</i>	Radius of bob d/2	Total L	1	2	3	mean		
1									
2									
3									
4									
5									
6									

Plot L against T^2 , and then calculate the gradient of the straight line. Since $K = \frac{g}{4\pi^2}$, use it calculate a value for g .